

Jiahe Li

Newcastle Upon Tyne | 07763632620 | tonylijahe@icloud.com |

Junior Software Engineer with commercial experience designing, building, deploying, and maintaining full-stack internal applications in the marine sector. Experienced in **Python, Java, JavaScript/TypeScript, SQL**, React.js, backend APIs, database design, data analysis, and engineering software tools. Worked on applications for carbon-emissions tracking, vessel planning, powertrain simulation, and reporting dashboards. Also experienced with **Modelica**/OpenModelica for engineering modelling. Seeking a Software Engineer role focused on practical business applications, backend systems, or full-stack development.

Professional Work Experience

Junior Software Engineer | Newcastle Marine Service | Full Time | 04/2024 - Present

- Designed, built, deployed, and maintained full-stack internal applications supporting carbon-emissions tracking, vessel planning, powertrain analysis, and operational workflows.
- Developed tools for carbon-emissions tracking, vessel planning, data visualisation, and reporting.
- Cooperated with internal departments and external partners to gather requirements, deliver updates, and support project timelines.
- Built data analysis and visualization tools for other engineers.
- Experience with practical deployment workflows for internal business software.
- **Achievement:** Delivered and maintained internal engineering software across multiple iterations, supporting changing customer and stakeholder requirements.
- **Achievement:** Built and maintained **2 internal full-stack applications**, reducing manual workflow time by approximately **40%** through automation, data processing, and improved reporting workflows.

Relevant Business Project Experience

Advance Zero Emission Tool | Newcastle Marine Service | 04/2024 - Present

- Funded by **Innovate-UK** Clean Maritime Demonstration Competition.
- Developed and worked with Modelica-based simulation models for vessel powertrain analysis, including engine and battery-assisted operation.
- Created power train control logic aimed at reducing CO₂ emissions and improving fuel-consumption performance.
- Worked in a team to create an automated process for Modelica Model Simulation, data analysis and visualisation via dashboard.
- Technology used: **React.JS, Python, Modelica, Flask, ZMQ**.

Vessel Planner | Newcastle Marine Service | 11/2024 - Present

- Developed internal vessel planning tools through multiple feature iterations based on stakeholder requirements and feedback.
- Built frontend components for creating, editing, and managing vessel planning data.
- Implemented backend APIs for storing, retrieving, validating, and updating planning records.
- Designed relational database tables to support vessel planning workflows.
- Deployed the system on a VPS and maintained it after release, including bug fixes and feature updates.
- Worked across the full development lifecycle from initial design to production deployment and support.

- Technology used: **React.JS, Java, Spring Boot, PostgreSQL, Docker, Nginx.**

BIMCO Shipping KPI Tool | Newcastle Marine Service | 06/2025 - Present

- Developed application for fleet key performance indicator logging, calculating, and visualizing.
- Configured the application for deployment and maintenance in a Linux server environment.
- Managed environment variables, backend service configuration, and frontend/backend connectivity.
- Troubleshoot deployment issues including server processes, ports, application errors, and configuration problems.
- Released iterative updates after stakeholder feedback and maintained the application after deployment.
- Technology used: **React.JS, TypeScript, Express.js, SQLite, Prisma, Nginx, Additional Android prototype: Java, Android Studio.**

Education

NEWCASTLE UNIVERSITY: MSc COMPUTER SCIENCE 2022-2023 (MERIT)

NEWCASTLE UNIVERSITY: BSc (HONS) MARKETING AND MANAGEMENT 2019-2022

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, SQL.

Frontend: React.js, Vue.js, HTML, CSS, Vite

Backend: Flask, Django, Express.js, Spring Boot, MyBatis, REST APIs

Databases: PostgreSQL, SQLite, MySQL, MongoDB, Redis

DevOps & Deployment: Git, Linux, Docker, Nginx, TeamCity basics, tmux, VPS deployment

Engineering & Simulation: Modelica, OpenModelica, powertrain modelling

Data & Visualisation: Excel, Tableau, SPSS, Alteryx.

Other Activities

NEWCASTLE UNIVERSITY ARCHERY CLUB | 09/2022-07/2023

- Awarded Bronze Medal for Novice British Universities and Colleges Sports Indoor Archery Competition.

Interests and Additional Language Skills

- Archery
- Languages: Mandarin native; English professional working proficiency, IELTS 7.5 overall
- Hobbyist ROS2 robotics development.